LITERATURE REVIEW

Research Gaps in Multilingual RAG: Positioning a Citation-Grounded System for Private Indian-Language Archives

A structured review of seven recent papers on multilingual Retrieval-Augmented Generation, identifying the specific research space this project occupies and the questions it sets out to answer.

7 PAPERS REVIEWED | 4 RESEARCH QUESTIONS | 1 CONSOLIDATED GAP

SCOPE
Multilingual & Cross-Lingual RAG

FOCUS
Private English & Indian-Language Documents

PURPOSE
Justify the Project's Research Contribution

INTRODUCTION

Purpose of this Review

This review identifies the research space for a **resource-efficient, trustworthy RAG system** for private English and Indian-language document archives. Seven recent papers on multilingual and cross-lingual Retrieval-Augmented Generation are examined for what they contribute and, more importantly, for what they leave unresolved.

Each paper below is summarized by its **contribution** and its **gap** — the specific limitation that motivates a design decision in the proposed project. A **consolidated gap** and four **evidence-based research questions** then frame exactly what the project sets out to answer.

Papers reviewed

#	Paper	Year
1	Investigating Language Preference of Multilingual RAG Systems	2025
2	Resource-Efficient Multilingual RAG for Indian Languages	2026
3	M3-RAG: Unified Multimodal and Multilingual RAG	2025
4	Translation-Based Embedding Strategies for Non-English Documents	2024
5	A Multilingual Intelligent Document Processing System	2025
6	LegalRAG: A Hybrid RAG System for Multilingual Legal Information Retrieval	2025
7	Hybrid Layered Retrieval and Task-Aware Embeddings for Persian RAG	2026

LITERATURE

Paper-by-Paper Contributions & Gaps

1 Investigating Language Preference of Multilingual RAG Systems

Findings of ACL 2025 · 2025

CONTRIBUTION

Analyses a hidden problem in multilingual RAG: retrieval systems often favor high-resource languages or the language used in the question, and generative models may favor the query language or Latin-script output. Proposes **DKM-RAG**, which combines translated passages with the model's internal knowledge to reduce this language preference.

RESEARCH GAP

Does not focus on private OCR-derived enterprise documents, page-level provenance, or Indian-language deployment. It motivates evaluating **answer-language fidelity, Latin-script bias, and code-switching** in the proposed system.

2 Resource-Efficient Multilingual RAG for Indian Languages

Resource-Efficient Multilingual RAG · 2026

CONTRIBUTION

Uses a largely English-based embedding pipeline and translates only the final output, claiming support across **22 Indian languages** while reducing computational cost. Its central idea is that translating all content, or using expensive multilingual processing, may not always be necessary.

RESEARCH GAP

English-only retrieval can lose domain-specific legal or technical meaning. The unresolved question is **when direct multilingual retrieval, translation-based retrieval, or a hybrid route is best** for private multilingual documents — this paper does not test that boundary.

3 M3-RAG: Unified Multimodal and Multilingual RAG

M3-RAG · 2025

CONTRIBUTION

Addresses both language and image understanding together. It can retrieve text and images and uses a **routing mechanism** to decide whether a query needs text retrieval, image retrieval, or both — e.g. when a user's question requires a diagram or chart rather than only written text.

RESEARCH GAP

Does not establish trustworthy, page-cited QA for heterogeneous private PDF, DOCX, and scanned-document collections under limited hardware. **Tables and diagrams remain a future extension** for the proposed project rather than an implemented capability here.

4 Translation-Based Embedding Strategies for Non-English Documents

Embedding Strategy Study · 2024

CONTRIBUTION

Studies whether translating non-English queries or documents into English improves RAG retrieval, on the premise that English embedding models are often very strong. The limitation the authors note: translation adds delay and may distort specialized terminology, names, legal wording, or culturally specific concepts.

RESEARCH GAP

Does not provide an **adaptive policy** identifying which queries actually benefit from translation versus direct multilingual embeddings — leaving that decision unresolved for real deployments.

5 A Multilingual Intelligent Document Processing System

Document Processing System · 2025

CONTRIBUTION

Focuses on processing and querying multilingual documents, letting users upload files and ask questions across different languages — a practically important capability as organizations increasingly hold multilingual files and need a common way to access information.

RESEARCH GAP

A functional multilingual QA pipeline alone **does not demonstrate retrieval-method comparisons, source-faithfulness, out-of-scope behaviour, or robust evaluation** across document formats and languages — exactly the gap this project's evaluation plan targets.

6 LegalRAG: A Hybrid RAG System for Multilingual Legal Information Retrieval

LegalRAG · 2025

CONTRIBUTION

Develops a bilingual Bangla–English question-answering system for government police gazettes, improving on standard RAG with an additional LLM that checks retrieved-text relevance and refines the user query before generation. It outperforms a basic RAG system on its selected legal dataset.

RESEARCH GAP

Evaluation is **domain- and collection-specific**, and the authors report difficulty with out-of-context questions. This motivates **abstention and evidence-sufficiency checks**, plus broader multilingual private-document evaluation, in the proposed system.

Preprint: arxiv.org/abs/2504.16121

7 Hybrid Layered Retrieval and Task-Aware Embeddings for Efficient Persian RAG Systems

Persian RAG · 2026

CONTRIBUTION

Proposes a layered Persian RAG architecture combining FAQ retrieval, semantic vector search, BM25 keyword search, and an LLM fallback, plus **task-aware query processing** to align retrieval with the user's task. Combining semantic and keyword retrieval lets each catch information the other may miss.

RESEARCH GAP

The design is **Persian- and task/FAQ-specific**, and latency rises on limited GPUs. The proposed project adapts the hybrid idea to Indian languages while varying document type and hardware budget — conditions this paper does not test.

SYNTHESIS

Consolidated Gap

No selected study evaluates an end-to-end private-document RAG system that jointly compares direct multilingual, translation-based, and hybrid retrieval for English and Indian languages while simultaneously measuring citation correctness, answer-language fidelity, OCR robustness, abstention, latency, and cost.

FRAMING THE STUDY

Evidence-Based Research Questions

RQ1	Which retrieval strategy gives the best retrieval quality and cost for each language and document type?
RQ2	Does hybrid dense-plus-lexical retrieval and cross-encoder reranking improve page-cited answer faithfulness?
RQ3	Can evidence-sufficiency checks reduce unsupported answers without unacceptable latency?
RQ4	How consistently does the system answer in the requested language?

Together, these four questions operationalize the consolidated gap into measurable outcomes: RQ1 targets retrieval-strategy selection, RQ2 targets answer trustworthiness, RQ3 targets the abstention mechanism, and RQ4 targets multilingual consistency — matching the evaluation metrics defined in the project synopsis.